responses to events, but anything that you expect will take time should be done offline or handled outside of the event loop.

Where should data processing be done?

If your application needs to process something that you know will take a long time, then it simply has to be done outside the event loop. Send the processing somewhere else (to another program) with a callback so that it can be done asynchronously, with your program being notified when it's done.

How does the solution in listing 9.4 perform with a list of 1.5 million names? It doesn't block the event loop for 5 seconds, but it does take *minutes* to complete (try to execute it and go make yourself a coffee while you wait). Despite the slow performance, by treating the data source as an event emitter you're able to process the entire list without blocking the event loop. The same approach works in reverse—just as data can be treated as events, so can events be treated as data.

9.3.4. Events as data

Suppose now that the competitor data also contains the fastest time for each competitor on a particular ski run:

```
0212: Turnbill, Geralyn, '12:13'
0055: Spielvogel, Cierra, '11:55'
0072: Renyer, Connie, '11:33'
0011: Engholm, Ciara, '14:10'
```

You receive a call from Joe, who says that he wants to display only competitors who have a best time faster than 12:00. It's tempting to modify the existing code to add the condition:

```
for item in chunk
  insertAt = 0
  if scoreBetterThan item, '12:00'
    toInsert = { original: item, sortOn: lastName(item) }
```